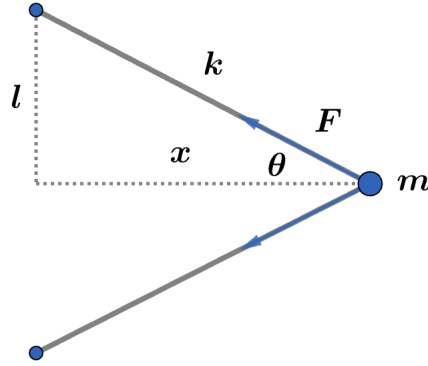


2016 F=ma Exam: Problem 7

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Suppose the springs have equilibrium length l and spring constant k . If we displace the mass m by distance x parallel to the spring, the spring force F is given by

$$F = k\Delta l = k(\sqrt{l^2 + x^2} - l)$$

Then the restoring force is

$$F_r = -2F \cos \theta = -2k(\sqrt{l^2 + x^2} - l) \frac{x}{\sqrt{l^2 + x^2}} = -2kx \left(1 - \frac{l}{\sqrt{l^2 + x^2}}\right)$$

For finite x , the magnitude of this restoring force is less than the simple harmonic force $F_{\text{SHM}} = -2kx$ so the actual acceleration of the mass is less than the assumed simple harmonic acceleration. Hence, the period T is longer than the SHM period

$$T_{\text{SHM}} = 2\pi \sqrt{\frac{m}{2k}}$$

As $x \rightarrow \infty$, $F_r \rightarrow F_{\text{SHM}}$ so $T \rightarrow T_{\text{SHM}}$. Since $T > T_{\text{SHM}}$ at small displacements, T decreases as x increases. Thus, the answer is C.